

Literature Review

History, Definition, and Prevalence of Cyberchondria

The origin of cyberchondria as a term is still debated with some attributing its first appearance to a 1999 Wall Street Journal and others to an article published in The Independent in 2001. The term, coined during the early years of the internet, was however not coined by a clinician or researcher (Starcevic et al., 2020). This term was first used to highlight the negative impact of technological advances due to the internet. The term Cyberchondria is made up of two words - ‘cyber’ to describe internet use and ‘hypochondria’ to describe health anxiety (Jungmann & Dessauer, 2025; Starcevic & Berle, 2013). Arriving on a consensus regarding the definition of cyberchondria has been challenging. There are two approaches to defining it. The first approach suggests linking cyberchondria to health anxiety further describing it as an excessive pattern of searching for health-related information that can cause subsequent increase in health anxiety and distress. The second approach suggests that “cyberchondria is syndrome-like and multidimensional in that it reflects both anxiety and an element of compulsiveness” (Starcevic et al., 2020, p.56).

A Scoping Review of 42 studies estimated the prevalence rates of cyberchondria to range from 30.7% to 55.6%. The prevalence rates varied across populations. Information sector employees in India had the highest prevalence of cyberchondria. First year medical students in Indonesia had a prevalence rate of 37.65% whereas Canada and the USA had a prevalence rate of 30.7% (Miezah et al., 2025).

Theoretical Models and Mechanisms of Cyberchondria

Cognitive-Behavioral Model

Initial conceptualization of cyberchondria emerged from the Cognitive-Behavioral

Theory which posits that cyberchondria is a manifestation of health anxiety that is driven by dysfunctional beliefs and misinterpretations of bodily sensations. High level of health-related attentiveness and low tolerance for ambiguity leads to catastrophization of symptoms. This catastrophization leads to excess reassurance seeking for relief (Martino et al., 2026). The reassurance-seeking model by Starcevic and Berle (2013) suggests that individuals with health anxiety engage in online health-related search to seek reassurance for their health concerns. Individuals who do not feel reassured continue searching for health-related information online in an effort to seek reassurance. This may continue despite frequent disappointments and distress following online searches. On the other hand, individuals who find reassurance from their first online search may go again to seek more reassuring information. 75% of internet users confuse the rankings of health-related search results with the probability of that disease. A less likely result that suggests a serious health condition will be taken into account more than a more likely benign condition. This might increase anxiety and lead to further online searches. This persisting search for health information is a maladaptive way of coping with health anxiety. Online health research may also persist due to various factors that reinforce it, such as credibility of information available, need for definitive explanations and information overload.

Metacognitive Model

The metacognitive model of cyberchondria links metacognitive beliefs to cyberchondria. According to this model, online health research and associated anxiety and depression are related to specific metacognitive beliefs (Starcevic et al., 2020). The metacognitive model of cyberchondria, grounded in the Self-Regulatory Executive Function (S-REF) model, posits that multiple metacognitive factors are involved in the development

and maintenance of psychological disorders (Spada et al., 2008). Beliefs about thinking (metacognitive beliefs), more than beliefs about other things such as illness or physical symptoms, are linked to psychological disturbance (Bailey & Wells, 2015). This is underpinned by the Cognitive Attentional Syndrome (CAS), a pattern of information processing defined by worry, rumination and monitoring of threat (Keen et al., 2022). The CAS is linked to dysfunctional metacognitive beliefs, also including beliefs and plans to regulate cognition (Wells, 2019). These metacognitions can be of two types- positive and negative. Positive metacognitive beliefs reflect the benefits of controllability of thinking (e.g., worrying about danger will mean I'm always prepared) for coping with anxiety whereas negative metacognitive beliefs (e.g., my thoughts will cause harm) consist of uncontrollability or dangers of thinking (Fergus & Spada, 2018; Sellers et al., 2016).

Prominent negative metacognitive beliefs result in “compulsive online health research” such that the search for information becomes threatening, compulsive, and distressing (Starcevic et al., 2020). Bailey and Wells (2015, as cited in Fergus & Spada, 2018) expounded on three sets of metacognitive beliefs related to health-related thoughts namely biased thinking beliefs (e.g., “Worrying about my health will help me cope”), thought-illness fusion beliefs (e.g., “Worrying about illness is likely to make it happen”), and beliefs about the uncontrollability of thoughts (e.g., “Dwelling on thoughts of illness is uncontrollable”) (Fergus & Spada, 2018, p. 4). Positive associations were found between these three health-related metacognitive beliefs and cyberchondria.

Digital Ecosystem and Generative AI Systems

Cyberchondria and the internet are inseparable. When searching for health-related information online, individuals may enter vague symptoms. Algorithms may surface high

engagement contents or sensational content which increases perceived threat and distress.

Generative AI symptom checkers may increase anxiety by suggesting rare or severe medical diagnoses (Martino et al., 2026).

Neuropsychological Mechanisms

Cognitive distortions and neuropsychological vulnerabilities influence how individuals perceive and interpret health-related information. Confirmation bias refers to the tendency to favour information that supports an individual's fears even when presented with contradictory evidence. While searching for information online, individuals may contemplate the worst case scenarios. Due to availability heuristic, individuals may disproportionately recall very serious medical conditions, overestimating their prevalence (Martino et al., 2026). Another neuropsychological condition at play is selective attention where individuals with health anxiety may engage in attentional bias towards bodily sensations and health-related information. Because of this, individuals may end up catastrophizing their bodily sensations (Martino et al., 2026).

Measures of Cyberchondria

Even though several tools have been developed to measure cyberchondria, the primary and most reliable tool used is the Cyberchondria Severity Scale (CSS). McElroy and Shevlin (2014) developed a 33-item multidimensional Cyberchondria Severity Scale (CSS-33) to assess compulsion, distress, excessiveness, reassurance seeking, and mistrust of medical professional. The CSS exhibited good psychometric properties with high internal consistency among the subscales as well as good concurrent and convergent validity. Fergus (2014) examined the psychometric properties of the CSS on a large sample of community adults in the United States. Confirmatory Factor Analysis (CFA) supported the five factor structure of

the scale. However, a higher order CFA revealed the ‘mistrust of medical professional’ subscale to be a distinct construct from the other subscales (Fergus, 2019).

A 15 item shorter version of CSS was developed on a German sample. CSS-15 demonstrated good internal consistency and split-half reliability and the confirmatory factor analysis confirmed the five factor structure (Barke et al., 2016). A 12 item short version of CSS was developed removing the ‘mistrust of medical professional subscale’. 12 items from four subscales (excluding mistrust) were retained. Confirmatory Factor Analysis was conducted to examine the validity of the scale. Stronger correlations were observed between CSS-12 and Short Health Anxiety Inventory (SHAI) than GAD-7 (McElroy et al., 2019).

Several other scales have also been developed to measure cyberchondria. A 27 item Cyberchondria Scale (CS), a multidimensional, self-report, culture specific scale was developed by Batigun et al (2018). The scale was found to be a reliable and valid measure to be used in clinical and health psychology studies in Turkey. Jokic-Begic et al. (2019) developed and validated a four item Short Cyberchondria Scale (SCS). The scale exhibited sufficient reliability and validity. The Brief Cyberchondria Scale (BCS) is a ten item measure of Cyberchondria developed in Puerto Rican and Caribbean. The BCS demonstrated good internal consistency and convergent validity (Gonzalez-Rivera, 2020).

Predictors and Correlates of Cyberchondria

A systematic review conducted by Miezah and colleagues (2026) focused on various correlates and predictors of cyberchondria. Positive correlations were found between internet literacy, smartphone addiction and problematic use of internet. Among psychological predictors, health anxiety is a significant predictor of cyberchondria. Individuals with health anxiety may engage more in online health research behaviours to assuage their concerns.

Anxiety sensitivity and Intolerance of Uncertainty also predicted cyberchondria. Individuals with anxiety sensitivity are more likely to misinterpret their bodily sensations as signs of a serious medical condition.

Those with low tolerance for uncertainty might look for health-related information to reduce ambiguity (Miezah et al., 2026). Cyberchondria is also associated with compulsive internet use (Martino et al., 2026). Cyberchondria was also found to be positively correlated with obsessive-compulsive symptoms, particularly those related to contamination or washing, and responsibility for harm symptoms (Zheng et al., 2020). A negative correlation was found between general mental health and cyberchondria (Miezah et al., 2026). A study by Nasiri and colleagues (2022) focused on personality traits predicting Cyberchondria. Neuroticism and extraversion predispose individuals to cyberchondria through the mediating effect of cognitive bias, health-related metacognitions, and emotional dysregulation.

Gender differences in cyberchondria severity are inconsistent across several studies. Several studies have found higher cyberchondria levels in women as compared to men (Aleyeidi et al., 2025; Ambrosini et al., 2022; Jungmann & Dessauer, 2025; Ozbas et al., 2025; Santoro et al., 2025). The meta analysis by Miezah et al. found that female students were more likely to search for health information online after discovering unexplained symptoms in their bodies and upon finding troubling information they would visit medical professionals. Dogan and Oksas (2025) found significantly higher cyberchondria levels in men as compared to women. Few studies also found no significant difference in cyberchondria scores between men and women (Aslantas & Altuntas, 2023; Bahadir & Dundar, 2024). Cyberchondria was also found to be more prevalent among young adults, particularly those under 35 years (Aslantas & Altuntas, 2023; Bahadir & Dundar, 2024). This

can be attributed to increased familiarity with technology and the internet that the young individuals have. However, some studies have also reported that there were no significant differences in age and cyberchondria levels (Dogan & Oksas, 2025; Ozbas et al., 2025)

Role of Metacognitive Beliefs

The Metacognitive model, based on the Self-Regulatory Executive Function (S-REF) theory posits that metacognitive beliefs may be more predictive of psychopathology than cognitions. Negative metacognitive belief about uncontrollability and danger of worry predicts health anxiety more than somatosensory amplification, neuroticism, and illness cognition (Bailey & Wells, 2013). A study by Melli et al. (2016) revealed that metacognitive beliefs about uncontrollability and interference of illness predicted health anxiety more than depression, general anxiety, anxiety-sensitivity, and health-related dysfunctional beliefs. Additionally, moderation analysis revealed that these metacognitive beliefs moderated the relationship between anxiety sensitivity and health anxiety.

As established before, cyberchondria is a phenomenon where individuals experience increased health anxiety due to repeated online searches of health-related information (Boussaid et al., 2025). Cyberchondria has been associated with metacognitive dysfunctions, particularly beliefs regarding uncontrollability and dangers of thinking (Martino et al., 2026). Metacognitive belief was found to be correlated to, as well as, significantly predicted cyberchondria (Nadeem et al., 2022). Airolidi et al. (2020) in their study found that metacognitions about uncontrollability of thoughts and metacognitions about biased thinking positively correlated with cyberchondria. Metacognitive beliefs significantly mediated the relationship between health anxiety and cyberchondria. Fergus and Spada (2018) tested a metacognitive conceptualization of cyberchondria including metacognitive beliefs concerning

health-related thoughts, beliefs about rituals and, stop signals. The study found that metacognitive beliefs about uncontrollability of health-related thoughts were found to be the most relevant to cyberchondria. Women showed stronger metacognition-cyberchondria links (Shahani et al., 2023). Despite consistent predictive power, longitudinal and intervention studies remain very limited.

Role of Anxiety

Unlike several constructs of anxiety like health anxiety, anxiety sensitivity and obsessive-compulsive symptoms, studies on cyberchondria-anxiety link are mixed and limited. Gioia and Boursier (2020) found a positive correlation between anxiety and cyberchondria on a sample of Italian women. Unemployed women seemed to show higher anxiety and cyberchondria symptoms as per the study findings. Similar findings were observed in other studies.

Cyberchondria was found to positively correlate with different subjective experiences of anxiety including COVID-19 anxiety, general anxiety, health-anxiety, and metacognitive beliefs about anxiety (Vismara et al., 2021). Some studies treated anxiety as an outcome of cyberchondria, not just a predictor. Ambrosini et al. (2022) found that cyberchondria directly predicted generalized anxiety. The study found that internet addiction and obsessive-compulsive symptoms explain the mechanism that links cyberchondria to anxiety and health anxiety. Contrasting evidence was found in a study by Arsenakis and colleagues (2021). Anxiety did not significantly predict cyberchondria when included alongside other variables such as health anxiety, obsessive-compulsive symptoms, intolerance of uncertainty, depression, and somatic symptoms. The study concluded that forms of anxiety that are not related to an individual's health may not be associated with cyberchondria. The reason might

be that people with high levels of anxiety are eager to search online for their symptoms to experience relief from anxiety. However, due to the vast amount of information available online, people may misinterpret their symptoms.

Relationship between Metacognitive Beliefs and Anxiety

According to the metacognitive theory, increased positive metacognitive beliefs (benefits of worry) combined with negative metacognitive beliefs (uncontrollability of thinking) predicts anxiety. Ryum et al. (2017) found that worry and metacognitive beliefs predicted anxiety independent of each other. It was also found that negative metacognitive beliefs were found to be particularly important in the development of anxiety through analyses of MCQ-30 subscales. A meta-analysis of 40 papers revealed that all five domains of metacognitive beliefs (positive beliefs about worry, negative beliefs about uncontrollability and danger of worry, cognitive confidence, need for control, and cognitive self-consciousness) significantly correlated with symptoms of anxiety (Thingbak et al, 2024).

Taken together, the reviewed literature demonstrates that cyberchondria is not just shaped by excessive internet use but also cognitive, emotional and metacognitive processes. Individuals engage in online health-related searches as a form of reassurance seeking to reduce distress associated with threatening bodily sensations. However, ambiguous and threatening health information can increase anxiety and distress instead of reducing it, reinforcing further internet searches. The metacognitive model suggests that individuals' beliefs about their own thoughts can play an important role in maintaining cyberchondria. Positive beliefs such as worrying is useful can increase online health research while negative beliefs about uncontrollability of thinking can increase distress.

Indian Studies

Literature evidence so far shows that cyberchondria is not just a by-product of the internet. It is a phenomenon shaped by maladaptive metacognitive beliefs and anxiety related vulnerabilities. However, a bulk of information comes from western studies. It is therefore necessary to study cyberchondria patterns within the Indian context.

India reports one of the highest Cyberchondria prevalence rates (Miezah et al., 2026). A prevalence rate of 55.6% was found in a study on IT sector employees (Makarla et al., 2019). The study also found general mental health was negatively correlated to cyberchondria severity. Vemuri and Prusty (2025) found that North India has the highest cyberchondria rates whereas south India has the lowest rates. Gandla et al. (2021) found high cyberchondria scores in their sample. Participants with previously diagnosed illnesses exhibited higher cyberchondria levels. The study found no significant gender differences in cyberchondria scores. On the contrary, significant gender differences in cyberchondria were found in another study on a sample of participants from Odisha, India. Females showed greater cyberchondria scores as compared to males (Ojha, 2025). Overall, the available Indian studies make a substantial contribution to the cyberchondria literature by demonstrating both the prevalence and contextual diversity of the phenomenon.

Despite a growing body of research on cyberchondria, several gaps remain that limit theoretical and practical advancement. First, much of the existing literature focused on health anxiety, anxiety sensitivity, intolerance of uncertainty, and obsessive- compulsive symptoms. Comparatively fewer studies have investigated the role of general anxiety in maintaining or precipitating cyberchondria. Existing literature also show inconsistent findings, with some studies finding significant positive correlations between general anxiety and cyberchondria

and others finding anxiety to be a non-significant predictor of cyberchondria. Second, despite the growing number of studies on cyberchondria, a causal relationship is yet to be determined. Third, existing literature on cyberchondria is dominated by western studies, limiting cultural generalizability of these results. Indian studies on cyberchondria largely focus on prevalence of cyberchondria, its correlates and demographic differences. Studies have not focused on the underlying mechanism of metacognitive beliefs on cyberchondria in an Indian sample.

The present study attempts to address these gaps by examining the mediating effect of metacognitive beliefs on cyberchondria and general anxiety on young Indian adults as well as studying gender differences in cyberchondria severity.